

**EDUCATION**

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**The University of Chicago, Chicago, IL**

*Bachelor of Science in Physics, Expected June 2028*

*Bachelor of Science in Molecular Engineering, Expected June 2028*

*- 4+1 Scholars Program, Masters of Engineering in Quantum Engineering, Expected June 2029*

Relevant Coursework: QuantumLab; Intermediate Quantum Engineering (Currently Enrolled); Machine Learning and Artificial Intelligence for Molecular Discovery and Engineering (Currently Enrolled); Principles of Engineering Analysis I-II; Molecular Engineering Transport Phenomena (Currently Enrolled)

**RELEVANT EXPERIENCE**

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**Grainger, Lake Forest, IL**

Data Science Apprentice | Jun 2026 - Aug 2026

- Processed and structured large multi-variable datasets in SQL and R, building reusable pipelines to prepare data for ML analysis
- Applied k-means clustering, PCA, and Firth logistic regression to large-scale behavioral datasets to identify patterns and predict outcomes for the customer base
- Built a Python-based data visualization pipeline (Streamlit) to surface trends in large multi-variable datasets, enabling faster pattern recognition

**Self-Learning Robotic Epitaxy Project | Yang Laboratory, University of Chicago**

Undergraduate Research Assistant | Feb 2026 - Present

- Contribute to the development of AI-driven robotic molecular beam epitaxy (MBE) system for manufacturing superconducting quantum materials and devices, combining materials science with machine learning
- Developed a Python script to enhance analysis of RHEED images by mapping pixel intensity to a false-color spectrum, improving pattern visibility for MBE process evaluation
- Engineered a Python computer-vision pipeline for automated RHEED analysis of 3,000+ frame MBE growth sequences, integrating VGG16 feature extraction, PCA/Ward clustering, NMF, and changepoint detection to identify evolving surface states and statistically characterize growth transitions.

**Physics Research | Hunt Laboratory, University of Chicago**

Undergraduate Researcher | Feb 2026 - Present

- Designed and built an embedded control system using an Arduino microcontroller to drive a stepper motor with precision, gaining hands-on experience in hardware programming and electromechanical systems
- Engineered and modeled a hydrodynamic device that generates surface waves through rotational motion, exploring fluid-structure interaction principles with applications in wave energy and physical simulation

**PROJECTS**

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**Quantum Machine Learning (QML) Project – UChicago Quantum Society, Chicago, IL**

- Co-authored a technical introduction to QML bridging quantum computing and machine learning, contributing sections on the Quantum Approximate Optimization Algorithm, hybrid quantum-classical algorithms, and barren plateaus in optimization
- Participated in a student-run organization dedicated to advancing understanding of quantum science and technology through lectures, workshops, and research engagement

**Controlled Wave Generator | Hunt Lab, University of Chicago**

- Modeled a floating wave-generating device in Fusion 360 and fabricated it using a resin 3D printer
- Programmed an ATtiny85 microcontroller to drive a DC motor at a controlled RPM to produce repeatable wave patterns
- Utilized a color wheel beneath a water-filled tub to enhance wave visibility for outreach demonstrations

**EL Enrollment and Non-EL Discipline Rates Analysis | University of Chicago**

- Merged and analyzed CRDC and CCD administrative datasets in R, building an analytic sample of 180,490 school-year observations across four years
- Applied OLS regression (linear, quadratic, fixed-effects) to examine the association between EL enrollment share and non-EL suspension rates, controlling for poverty and school size
- Presented findings via an in-class poster presentation and a formal research report

**AWARDS & HONORS**

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|------------------------------------------------------------------|-----------------------------------------|
| • <b>Hispanic Scholarship Fund Scholar</b> (2024)                | • <b>UChicago Dean’s Scholar</b> (2024) |
| • <b>United States Marine Corps Scholastic Excellence</b> (2024) | • <b>Illinois State Scholar</b> (2024)  |

**SKILLS**

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|-----------------------------|----------------------------------------|----------------------------|
| <b>Languages</b>            | <b>Technical</b>                       | <b>Tools</b>               |
| • Conversational in Spanish | • Java, Python, R, MATLAB, SQL, Fusion | • Google Workspace & Excel |